

Experimental Phase Equilibria Studies of the Pb-Fe-O System in Air, in Equilibrium with Metallic Lead and at Intermediate Oxygen Potentials



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The phase equilibria information on the Pb-Fe-O system is of practical importance for the improvement of the existing thermodynamic database of lead-containing slag systems (Pb-Zn-Fe-Cu-Si-Ca-Al-Mg-O). Phase equilibria of the Pb-Fe-O system have been investigated: (a) in air at temperatures between 1053 K and 1373 K (780 °C and 1100 °C); (b) in equilibrium with metallic lead at temperatures between 1053 K and 1373 K (780 °C and 1100 °C); and (c) at intermediate oxidation conditions for the liquid slag in equilibrium with two solids (spinel + magnetoplumbite), at temperatures between 1093 K and 1373 K (820 °C and 1100 °C). The high-temperature equilibration/quenching/electron probe X-ray microanalysis technique has been used to accurately determine the compositions of the phases in equilibrium in the system. The Pb and Fe concentrations in the phases were determined directly; preliminary thermodynamic modeling with FactSage was used to estimate the ferrous-to-ferric ratios and to present the results in the ternary diagram.

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I. INTRODUCTION

MOST industrial lead smelting operations include a process step that involves the reduction of lead slag and the separation of metallic lead bullion. These of high-PbO slags also contain iron oxide, SiO₂, CaO, ZnO, and other components, such as “Cu₂O,” Al₂O₃, and MgO. To optimize the conditions of these industrial processes, detailed information is required of: (a) slag–solid equilibria at various oxygen potentials; (b) the activities of lead species in the slag at corresponding conditions; and (c) the partitioning of elements between metal, slag, solids, and gas phases.

The present study is a part of an overall integrated thermodynamic modeling and experimental research program to support the optimization and development of complex lead smelting, refining, and recycling processes. An essential part of the program is the experimental determination of gas–slag–metal–solid oxide equilibria in those Pb-containing systems where the appropriate information is lacking or controversial. This particularly applies to all iron-containing slag systems in equilibrium with metallic lead, since the previous

research^[1,2] has not covered these conditions. The lead-containing slag systems studied previously^[1–11] were focused on equilibria in air, with limited investigations^[12] at most reducing conditions (equilibria with metallic iron), which have low PbO concentrations in the slag (<1 wt pct PbO). Only one study^[13] was found on the Pb-Fe slag in equilibrium with Pb metal, limited to a single 1473 K (1200 °C) isotherm at tridymite saturation in the Pb-Fe-Si-O system.

The Pb-Fe-O is one of the low-order binary and ternary systems essential for the development of an accurate thermodynamic database for the multi-component Pb-Zn-Fe-Cu-Si-Ca-Al-Mg-O system. The Pb-Fe-O system is also important for the characterization of lead-assisted corrosion of structural steels in nuclear power plants.^[14]

At high temperatures, lead and iron oxides form a complete range of liquid solutions with a eutectic at the PbO-rich side. Despite numerous attempts to identify the stable solid phases in this system,^[15–31] there are still significant uncertainties in their compositions and temperature ranges of stability. It has been found as follows:

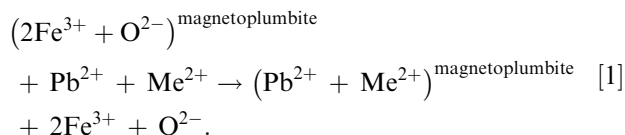
1. There are three stable compounds with compositions close to the PbO-FeO_{1.5} join, including hexagonal magnetoplumbite $\text{Pb}_{1+x}\text{Fe}_{12-x}\text{O}_{19-x}$ ($\text{P}_{1+x}\text{F}_{12-x}$), hexagonal plumboferrite $\text{PbFe}_{5+x}\text{O}_{8.5+1.5x}$ (PF_{5+x}), and tetragonal perovskite-based lead ferrite compound $\text{Pb}_2\text{Fe}_{2\pm x}\text{O}_5$ ($\text{PF}_{1\pm x}$).^[32,33] (Note: the abbreviations “P” for PbO and “F” for FeO_{1.5} or FeO will be used in this paper.)

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2. All these compounds melt incongruently. Some of them appear to decompose when slowly cooled to room temperature.^[34] Magnetoplumbite $Pb_{1+x}Fe_{12-x}$ is stable in air up to ~1573 K to 1623 K (1300 °C to 1350 °C), and is often present as a primary phase on crystallization of industrial lead slags.^[8–10] It is also an important component of a so-called lead sludge,^[35] a by-product of some lead waste recycling techniques.
3. Some or all of these compounds have significant non-stoichiometry along the PbO –“ $FeO_{1.5}$ ” section.
4. As many as eleven compounds have been reported in the literature, with the (P/F = $PbO/FeO_{1.5}$) ratios equal to 4:1, 3:2, 1:1, 3:4, 1:2, 1:4, 1:5, 1:6, 1:8, 1:10, and 1:12.^[36] Many of these compounds are metastable at all conditions, or belong to the solid solution ranges of the three main compounds listed above.

In addition, Holtstam^[37–39] studied several phases that can take divalent cations into solutions: lindqvistite,^[37] $Pb_2MeFe_{16}O_{27}$ ($Me = Mn^{2+}, Mg^{2+}$); in naturally formed plumbogerrite^[38]: $Pb_2Me_{0.33}Fe_{10.67}O_{18.33}$ ($Me = Mg^{2+}, Mn^{2+}$, and possibly Fe^{2+}); and magnetoplumbite^[39]: $Pb_{1+x}Me_xFe_{12-2x}O_{19-x}$ ($Me = Mn^{2+}, Zn^{2+}$, and possibly Fe^{2+}). In the case of plumbogerrite with complete Me substitution by Fe^{2+} , the atomic ratio would be equal to $Fe:Pb = 5.5$, and only 1/33 of total iron (3 pct) would be in the ferrous state. Much greater values of substitution were observed for magnetoplumbite, where the main mechanism of substitution by a divalent metal ($Me = Fe, Zn, Mn$) is suggested to be^[39]



During the development of the present public FactSage thermodynamic database,^[40] Jak^[2] used mainly the data on the solid phase compositions by Nevrya,^[27] which were among the most contemporary for that time. These compositions, $Pb_2Fe_2O_5 = \text{“PF,”}$ $PbFe_4O_7 = \text{“PF}_4\text{,”}$ and $PbFe_{10}O_{16} = \text{“PF}_{10},\text{”}$ are now present in the FactSage public database as stoichiometric compounds.

However, several new accurate experimental studies using improved analytical methods have been published since this database was developed (Diop, 2010^[30] and Sahu, 2012^[36]), reporting the $PbFe_5O_{8.5}$ – $PbFe_6O_{10}$ (PF_5 – PF_6) range of solid solution for plumbogerrite, and $PbFe_{12}O_{19} = \text{“PF}_{12}\text{”}$ instead of “ PF_{10} ” for magnetoplumbite.

The current study focuses on the phase equilibria of the systems: (a) gas–slag–solid in air atmosphere at temperatures between 1053 K and 1373 K (780 °C and 1100 °C); (b) slag–solid oxide–liquid lead metal at temperatures between 1053 K and 1373 K (780 °C and 1100 °C); and (c) slag–spinel–magnetoplumbite equilibria at intermediate oxidation conditions, at temperatures between 1093 K and 1373 K (820 °C and 1100 °C). In

the (b) and (c) cases, the system was studied under closed conditions (high-purity argon was used as an atmosphere). The upper temperature of 1373 K (1100 °C) was limited by the ability to quench the slag phase as a homogeneous liquid and the vapor pressure of Pb and PbO .

II. EXPERIMENTAL TECHNIQUE AND PROCEDURE

A. Experimental Procedure and Sample Examination

The experimental procedure and apparatus were described in detail in previous publications.^[41] In brief, synthetic samples were equilibrated at controlled conditions, quenched, and the compositions of the resulting phases were measured using electron probe X-ray microanalysis (EPMA).

Initial mixtures were prepared by mixing high-purity powders of Fe_2O_3 (99.945 wt pct purity), PbO (99.9 wt pct purity), supplied by Alfa Aesar, MA; and “ FeO ” (99.9 wt pct purity, supplied by Sigma Aldrich, MO). The actual formula for the reagent “ FeO ” was assumed to be $FeO_{1.05}$, because no stable phase of exact FeO composition exists. For experiments in equilibrium with metallic lead, Pb powder (99.95 wt pct purity; supplied by Sigma Aldrich, MO) was added, 20 to 30 wt pct in excess of the mass of the oxide mixture, to ensure that metallic lead was present at the final equilibrium. The initial compositions of the mixtures were selected so that one or more crystalline phases would be present in equilibrium with liquid slag. The volume fraction of solids was targeted to be below 50 pct, and preferably approximately 10 pct, to achieve best conditions for maintaining the homogeneity of the highly fluid slag on quenching. Less than 0.3 g of mixture was used in each equilibration experiment.

Five types of substrates were used for equilibration, depending on the conditions:

1. Magnetite (spinel) baskets, prepared by heating a folded iron foil (99.5 wt pct purity, 0.1 mm thickness, supplied by Sigma Aldrich, MO) in CO_2 at 1473 K (1200 °C) for 2 to 3 hours (the stoichiometry of Fe_3O_4 was checked gravimetrically), were used for the experiments where spinel was one of the primary phases (slag + spinel + Pb metal and slag + spinel + magnetoplumbite).
2. Hematite baskets, prepared from the same iron foil as above by heating in air at 1473 K (1200 °C) for 2 to 3 hours (the stoichiometry of Fe_2O_3 was checked gravimetrically), were used for the (slag + lead ferrite) experiments in air at 1053 K to 1273 K (780 °C to 950 °C). Hematite is not stable in equilibrium with slags at these conditions, and reacts with the formation of a layer of lead ferrites ($Pb_{1+x}Fe_{12-x}$, PF_{5+x} or $PF_{1\pm x}$) that covers the surface of the substrate and prevents further reaction with the liquid due to the slow solid-state diffusion.
3. Platinum or platinum-rhodium wires, bent in spiral shape to hold the sample by means of wetting and surface tension, were used for the (slag + magneto-

plumbite) experiments in air at 1273 K to 1373 K (1000 °C to 1100 °C). Later pure platinum was found to be preferable, since rhodium was less resistant in oxidizing conditions and may dissolve in slag as an oxide (up to 1 wt pct Rh_2O_3) or also precipitate as $\text{Pb}_3\text{Rh}_7\text{O}_{15}$ crystals. However, even pure platinum may react with PbO-rich slag at low temperatures in air, forming the compounds PbPt_2O_4 and Pb_2PtO_4 .^[42] Any unwanted reactions were checked by EPMA measurements, and only results of the uncontaminated samples were used.

4. Gold baskets or wires bent in spiral shape were used for experiments in the PbO , $\text{PF}_{1\pm x}$, and PF_{5+x} primary phase fields in air at 1053 K to 1148 K (780 °C to 875 °C). Gold has been found to be the most resistant to attack by high-PbO slags in oxidizing conditions.
5. Iridium L-shaped plates, provided by Furuya Metal, Japan, were used for equilibration of slags with Pb metal and solids other than magnetite—*i.e.*, $\text{PF}_{1\pm x}$ or PbO , at 1053 K to 1103 K (780 °C to 830 °C). Iridium is the only refractory metal stable against the combination of molten Pb and PbO ,^[43] though it cannot be used in oxidizing conditions without Pb metal: it reacts as $\text{Ir} + \text{PbO} + \text{O}_2 = \text{PbIrO}_3$ (or $\text{Pb}_2\text{Ir}_2\text{O}_{7-x}$).

All equilibration experiments were carried out in a vertical reaction tube (impervious recrystallized alumina, 30-mm inner diameter) using an electrical resistance silicon carbide (SiC) heated furnace. A working thermocouple in a recrystallized alumina sheath was placed immediately next to the sample to monitor the actual sample temperature. The working thermocouple was periodically calibrated against a standard thermocouple (supplied by the National Measurement Institute of Australia, NSW, Australia). The overall absolute temperature accuracy of the experiment was estimated to be ± 3 K. An open system reaction tube was used for equilibration of samples with air. In the case of experiments at lower oxygen potentials, an inert atmosphere of high-purity Argon gas (99.999 pct purity; supplied by Coregas, NSW, Australia) in a gas-tight system was used; the oxygen partial pressures were defined by equilibria of three condensed phases (liquid slag + two solid oxides or liquid slag + liquid metal + solid oxide), but they were not measured directly. Samples were suspended in the hot zone of the furnace by Kanthal support wire (0.7 or 1-mm diameter), pre-melted for 30 minutes at 20 K to 130 K (-253 °C to -143 °C) above the equilibration temperature, and then equilibrated at the final target temperature and atmosphere condition for required time. At the end of equilibration process, the sample was rapidly quenched in iced calcium chloride brine [<255 K (-18 °C)]. The specimen was then washed thoroughly in water and ethanol before being dried on a hot plate, mounted in epoxy resin, and cross-sectioned using conventional metallographic polishing technique.

The samples were examined by optical microscopy, carbon-coated and the phase compositions were measured using an electron probe X-ray microanalysis

technique with wavelength-dispersive detectors (JEOL 8200L EPMA; Japan Electron Optics Ltd., Tokyo, Japan). The EPMA was operated with 15-kV accelerating voltage and 20 nA probe current using the Duncumb–Philibert atomic number, absorption, and fluorescence correction (ZAF correction). Fe_2O_3 (supplied by Charles M. Taylor Co., Stanford, CA) and K-456 lead silicate glass (71 wt pct PbO , supplied by NIST) standards were used for calibration of the EPMA. The concentrations of metal cations were measured with EPMA; no information on the different oxidation states of the cations was obtained in this study; iron oxide concentrations were recalculated as either FeO or $\text{FeO}_{1.5}$ for presentation purposes. The molar ratios $\text{Pb}/(\text{Pb} + \text{Fe})$ were used for phase diagrams to avoid any ambiguity.

A homogeneous phase was formed on quenching of the slags containing 70 to 80 mol pct PbO at 1053 K to 1173 K (780 °C to 900 °C) (in lead ferrite or spinel primary phase fields). The compositions of liquid and the crystal phases were measured by EPMA-point analysis method. In the case of slags with lower PbO concentration (quenched from 1273 K to 1373 K (1000 °C to 1100 °C)) or higher PbO concentration (with PbO -massicot as a primary phase), the glassy homogeneous areas were limited to locations at the surfaces directly contacting the quenching medium. The approach for accurate, repeatable, and objective measurements by EPMA-line analysis similar to the one described by^[44] was used to obtain the average compositions of liquid phase. In many cases, it was found that very rapid quenching was required, and the experiments were repeated many times until an area of appropriate well-quenched microstructure was found.

B. Confirmation of Achievement of Equilibrium

To ensure the achievement of equilibrium in the samples, several measures were used including: (1) variation of equilibration time; (2) assessment of the compositional homogeneity of phases by EPMA; (3) approaching the final equilibrium point from different starting compositions; and, importantly, (4) consideration of specific reactions that may affect the achievement of equilibrium or reduce the accuracy. Several dedicated series of experiments were carried out for these purposes. For example, the equilibration of samples in the Pb-Fe-O system in equilibrium with metallic Pb can be approached using different starting mixtures: (i) from an “ $\text{FeO}_{1.05}$ ” + PbO mixture, reacting to form the Fe_2O_3 - FeO - PbO slag and liquid Pb metal; and (ii) from an Fe_2O_3 + Pb or Fe_2O_3 + Pb + PbO mixture, where some Pb reacts with Fe_2O_3 to form PbO and FeO as components of liquid slag. It was found that the initial mixtures of the type (i) provided more consistent results, with high proportion of homogeneous slag containing some suspended crystals of solid oxide(s) and liquid Pb droplets. Often, no areas with all three phases (slag + solid oxide + Pb metal) were present in the samples when the mixtures of the type (ii) were used, so those samples were rejected. This indicates that the reaction of

metallic Pb with Fe_2O_3 is kinetically slow, due to limited surface area of Pb drops that tend to coalesce. The opposite reaction, where Pb metal is gradually formed from the slag with excess of ferrous iron, is much faster. No difference was found in final composition of phases when started from the mixtures of the type (i) if the equilibration time is 1 hour or longer. Usually, a premelt of at least 1173 K (900 °C) was used to ensure liquid formation.

III. RESULTS AND DISCUSSION

Examples of micrographs of quenched PbO -“ FeO_x ” samples are given in Figures 1(a) through (d).

The results of experiments are reported in Table I. The phase diagrams constructed based on these results are given in Figures 2 through 4. The present results, in combination with previous data,^[15–33,36–39] were used to establish the limits of stability of three compounds in the Pb-Fe-O system: hexagonal magnetoplumbite $\text{Pb}_{1+x}\text{Fe}_{12-x}\text{O}_{19-x}$ ($\text{P}_{1+x}\text{F}_{12-x}$; Fe:Pb = 10 to 12 in air, 5 to 12 in all conditions), hexagonal plumboferrite $\text{PbFe}_{5+x}\text{O}_{8.5+1.5x}$ (PF_{5+x} ; Fe:Pb = 5 to 6), and tetragonal 1:1 lead ferrite $\text{Pb}_2\text{Fe}_{2\pm x}\text{O}_5$ ($\text{PF}_{1\pm x}$; Fe:Pb = 0.9 to 1.1), see Table II (note the abbreviations P = PbO, F = $\text{FeO}_{1.5}$ or FeO). Only the range of solid solubility of magnetoplumbite involves some ferrous iron; two other compounds are assumed to be on the PbO - $\text{FeO}_{1.5}$ join.

Preliminary thermodynamic modeling with FactSage was used to estimate the ferrous-to-ferric ratios in the liquid phase to construct a ternary phase diagram presented in Figure 5. A tentative isothermal section at 1148 K (875 °C) is given in Figure 6 to explain the phase relations involving the solid solutions in $\text{P}_{1+x}\text{F}_{12-x}$ (magnetoplumbite) and $\text{Pb}_x\text{Fe}_{3-x}\text{O}_4$ (spinel).

A. Liquidus in the PbO -“ $\text{FeO}_{1.5}$ ” System in Equilibrium with Air

The primary phases found in the PbO -“ $\text{FeO}_{1.5}$ ” system in equilibrium with air are PbO (massicot), $\text{PF}_{1\pm x} = \text{Pb}_2\text{Fe}_{2\pm x}\text{O}_{5\pm 1.5x}$, $\text{PF}_{5+x} = \text{PbFe}_{5+x}\text{O}_{8.5+1.5x}$, and $\text{P}_{1+x}\text{F}_{12-x} = \text{Pb}_{1+x}\text{Fe}_{12-x}\text{O}_{19-x}$ (see the diagram in Figure 2). The liquidus in the hematite (Fe_2O_3) and spinel (Fe_3O_4) primary phase fields at higher temperatures was not investigated in the present study. The ($\text{PF}_{1\pm x}$ + liquid) phase assemblage was observed in experiments from 1053 K to 1123 K (780 °C to 850 °C). No liquid was found at 1143 K (770 °C). Extrapolation of the PbO and $\text{PF}_{1\pm x}$ liquidus lines until their intersection gave the eutectic temperature $\text{Liquid} \rightarrow \text{PbO} + \text{PF}_{1\pm x}$ in air at 1045 ± 5 K (772 ± 5 °C, point F). Extrapolation of the PF_{5+x} and $\text{PF}_{1\pm x}$ liquidus lines until their intersection gave the peritectic temperature $\text{Liquid} + \text{PF}_{5+x} \rightarrow \text{PF}_{1\pm x}$ at approximately 1128 ± 5 K (855 ± 5 °C, point G) (see Figure 2). Undissolved PF_{5+x} crystals were found after 1 day of heating of the liquid + $\text{P}_{1+x}\text{F}_{12-x}$ samples at 1148 K (875 °C); the morphology of solids in the samples indicated that $\text{P}_{1+x}\text{F}_{12-x}$ was a stable phase, so the peritectic $\text{Liquid} + \text{P}_{1+x}\text{F}_{12-x} \rightarrow \text{PF}_{5+x}$ in air is

estimated to be slightly lower but close to 1148 K (875 °C), approximately at 1147 ± 5 K (874 ± 5 °C, point H). The temperature 1574 K (1301 °C, point J) for the peritectic $\text{Liquid} + \text{hematite} \rightarrow \text{P}_{1+x}\text{F}_{12-x}$ was taken from FactSage.^[40]

The observed solubility of iron oxide in PbO -massicot did not exceed 0.4 mol pct. Lead ferrite $\text{PF}_{1\pm x}$ solution was found to have a range of solubility 47 to 54 mol pct Pb/(Fe + Pb). Other authors^[33] also reported that this phase could have an excess of iron, with Fe:Pb ratio up to 8:7.

A range of Fe:Pb ratios between 5 and 6 was observed in the PF_{5+x} phase. No occurrences of “ PF_4 ” (PbFe_4O_7 , the stoichiometry accepted in the FACT53 database in FactSage^[40]) were found.

At 1148 K to 1373 K (875 °C to 1100 °C), the Fe:Pb ratio in the $\text{P}_{1+x}\text{F}_{12-x}$ crystals in equilibrium with liquid in air was always close to 10, with small deviations only, suggesting that plumboferrite PF_{5+x} and magnetoplumbite $\text{P}_{1+x}\text{F}_{12-x}$ exist as separate phases in oxidizing conditions.

B. Liquidus in the PbO -“ FeO_x ” System in Equilibrium with Metallic Pb

Most of the liquidus of the PbO -“ FeO_x ” system in equilibrium with metallic lead have either the Spinel ($\text{Pb}_x\text{Fe}_{3-x}\text{O}_4$, $0 < x < 0.1$, at 1073 K to 1373 K (800 °C to 1100 °C)) or the Massicot (PbO) as primary phases (Figure 3). At 1053 K and 1063 K (780 °C and 790 °C), slag + $\text{PF}_{1\pm x}$ + Pb metal was found to be in equilibrium. A peritectic reaction $\text{Liquid} + \text{spinel} \rightarrow \text{PF}_{1\pm x} + \text{Pb}$ is estimated to occur at approximately 1068 ± 5 K (795 ± 5 °C, point D). The liquid phase was found at 1053 K (780 °C), but not at 1043 K (770 °C); the eutectic ($\text{Liquid} \rightarrow \text{“PbO”} + \text{PF}_{1\pm x} + \text{Pb}$) is estimated to be approximately at 1044 ± 5 K (771 ± 5 °C, point E) (lower than the eutectic $\text{Liquid} \rightarrow \text{PbO} + \text{PF}_{1\pm x}$ in air, according to the Alkemade theorem). The temperature of this eutectic does not change significantly between the equilibrium with air and with Pb metal, which indicates that almost all iron is in the ferric state at both conditions. The solubility of iron in massicot (lead oxide) and $\text{PF}_{1\pm x}$ at metallic Pb saturation is similar to that observed in air. At 1043 K to 1123 K (770 °C to 850 °C), a solubility of about 1 to 2 mol pct PbO in spinel was observed in equilibrium with liquid or $\text{PF}_{1\pm x}$.

In several selected samples, the attempts to measure the composition of metallic Pb phase were undertaken. The measured Fe concentrations in Pb were found to be below the uncertainty limits for EPMA, estimated to be 0.2 wt pct (note that the actual Fe concentrations in lead were predicted to be $< 10^{-4}$ wt pct using FactSage^[40]).

C. Liquid-Spinel-Magnetoplumbite ($\text{P}_{1+x}\text{F}_{12-x}$) Equilibrium in the PbO -“ FeO_x ” System

The liquid-spinel-magnetoplumbite line of the Pb-Fe-O diagram is univariant and at a given temperature has fixed $p(\text{O}_2)$ controlled by the equilibrium between the condensed phases. Two special features are

Table I. Experimentally Measured Compositions of Phases in the PbO-FeO_x System at Various Oxygen Potentials

Final Equilibration T [K (°C)]	Substrate	Redox Condition	Premelt T [K (°C)]	Time (h)	Phase	Mol Pct PbO (Rest FeO _x)
Equilibrium with Pb metal						
Two solids						
1043 (770)	Fe ₃ O ₄ o.	Pb metal (in Ar)	1073 (900)	0.5 + 3	spinel PF _{1±x}	* 52.7
1043 (770)	Fe ₃ O ₄ o.	Pb metal (in Ar)	1073 (900)	0.5 + 3	PF _{1±x} “PbO”	56.6 99.5
1043 (770)	Fe ₃ O ₄ o.	Pb metal (in Ar)	1123 (850)	3 + 15	P _{1+x} F _{12-x} PF _{1±x}	15.5 52.7
1043 (770)	Fe ₃ O ₄ o.	Pb metal (in Ar)	1123 (850)	3 + 15	PF _{1±x} “PbO”	57.5 99.4
1053 (780)	Fe ₃ O ₄ o.	Pb metal (in Ar)	1123 (850)	1 + 5	spinel P _{1+x} F _{12-x}	2.3 15.7
1053 (780)	Fe ₃ O ₄ o.	Pb metal (in Ar)	1123 (850)	1 + 5	P _{1+x} F _{12-x} PF _{1±x}	15.7 51.8
1063 (790)	Ir plate	Pb metal (in Ar)	1113 (840)	1 + 2	spinel P _{1+x} F _{12-x}	2.2 15.2
1063 (790)	Ir plate	Pb metal (in Ar)	1113 (840)	1 + 2	P _{1+x} F _{12-x} PF _{1±x}	15.2 51.0
PbO (massicot) liquidus						
1063 (790)	Ir plate	Pb metal (in Ar)	1153 (880)	0.5 + 4	liquid “PbO”	78.9 99.88
1083 (810)	Ir plate	Pb metal (in Ar)	1143 (870)	0.5 + 4	liquid “PbO”	80.6 99.74
1103 (830)	Ir plate	Pb metal (in Ar)	1143 (870)	1 + 4	liquid “PbO”	85.9 99.88
PF (Pb ₂ Fe ₂ O ₅) liquidus						
1053 (780)	Fe ₃ O ₄ o.	Pb metal (in Ar)	1123 (850)	1 + 5	liquid PF _{1±x}	75.4 55.2
1053 (780)	Ir plate	Pb metal (in Ar)	1093 (820)	3 + 1.5	liquid PF _{1±x}	75.9 56.7
1063 (790)	Ir plate	Pb metal (in Ar)	1113 (840)	1 + 2	liquid PF _{1±x}	73.2 54.0
Spinel (Pb _x Fe _{3-x} O ₄ , 0 < x < 0.1) liquidus						
1073 (800)	Fe ₃ O ₄ o.	Pb metal (in Ar)	1123 (850)	0.5 + 2	liquid spinel	72.0 1.48
1123 (850)	Fe ₃ O ₄ o.	Pb metal (in Ar)	1173 (900)	1 + 2	liquid spinel	69.7 1.8
1173 (900)	Fe ₃ O ₄ o.	Pb metal (in Ar)	no	2	liquid spinel	67.9 *
1223 (950)	Fe ₃ O ₄ o.	Pb metal (in Ar)	1273 (1000)	0.5 + 2	liquid spinel	64.7 *
1273 (1000)	Fe ₃ O ₄ o.	Pb metal (in Ar)	1323 (1050)	2 + 1	liquid spinel	61.3 *
1323 (1050)	Fe ₃ O ₄ o.	Pb metal (in Ar)	1343 (1070)	1	liquid spinel	59.5 *
1373 (1100)	Fe ₃ O ₄ o.	Pb metal (in Ar)	1383 (1110)	1	liquid spinel	53.9 *
Spinel (Pb _x Fe _{3-x} O ₄) – PF _x (Pb _{1+x} Fe _{12-x} O _{10-x}) – slag						
1093 (820)	Fe ₃ O ₄ o.	intermediate p(O ₂) (in Ar)	1223 (950)	0.5 + 5	liquid spinel P _{1+x} F _{12-x}	69.4 2.6 16.1
1123 (850)	Fe ₃ O ₄ o.	intermediate p(O ₂) (in Ar)	1203 (930)	0.5 + 2	liquid spinel P _{1+x} F _{12-x}	68.4 * 15.5
1173 (900)	Fe ₃ O ₄ o.	intermediate p(O ₂) (in Ar)	1223 (950)	0.5 + 2.5	liquid spinel P _{1+x} F _{12-x}	64.7 * 15.1
1173 (900)	Fe ₃ O ₄ o.	intermediate p(O ₂) (in Ar)	1223 (950)	3	liquid spinel P _{1+x} F _{12-x}	65.4 2.5 15.7

Table I. Continued

Final Equilibration T [K (°C)]	Substrate	Redox Condition	Premelt T [K (°C)]	Time (h)	Phase	Mol Pct PbO (Rest FeO _x)
1223 (950)	Fe ₃ O ₄ o.	intermediate p(O ₂) (in Ar)	no	2.5	liquid spinel P _{1+x} F _{12-x}	61.2 3.0 13.4
1223 (950)	Fe ₃ O ₄ o.	intermediate p(O ₂) (in Ar)	1273 (1000)	0.5 + 4	liquid spinel P _{1+x} F _{12-x}	61.0 2.8 13.3
1273 (1000)	Fe ₃ O ₄ o.	intermediate p(O ₂) (in Ar)	1293 (1020)	0.5 + 1	liquid spinel P _{1+x} F _{12-x}	57.1 ~2.2* 12.3
1323 (1050)	Fe ₃ O ₄ o.	intermediate p(O ₂) (in Ar)	no	0.5 + 1	liquid spinel P _{1+x} F _{12-x}	53.5 2.4 11.7
1323 (1050)	Fe ₃ O ₄ o.	intermediate p(O ₂) (in Ar)	1373 (1100)	0.5 + 1	liquid spinel P _{1+x} F _{12-x}	53.7 2.7 11.4
1323 (1050)	Fe ₃ O ₄ o.	intermediate p(O ₂) (in Ar)	1373 (1100)	0.5 + 1	liquid spinel P _{1+x} F _{12-x}	53.1 2.7 11.3
1373 (1100)	Fe ₃ O ₄ o.	intermediate p(O ₂) (in Ar)	1393 (1120)	1	liquid spinel P _{1+x} F _{12-x}	49.9 2.4 10.5
Equilibrium in air						
Two solids						
1053 (780)	Fe ₂ O ₃ o.	in air	1173 (900)	1 + 3	PF _{5+x} PF _{1±x}	15.9 49.8
1073 (800)	Fe ₂ O ₃ o.	in air	1173 (900)	0.5 + 2	PF _{5+x} PF _{1±x}	15.2 50.8
1123 (850)	Fe ₂ O ₃ o.	in air	1173 (900)	0.2 + 1	PF _{5+x} PF _{1±x}	16.2 49.4
1123 (850)	Au o.	in air	no	48	PF _{5+x} PF _{1±x}	15.9 48.5
PbO (massicot) liquidus						
1083 (810)	Au wire	in air	no	0.5	liquid “PbO”	82.9 99.62
1053 (780)	Au wire	in air	no	4	liquid “PbO”	80.4 99.93
PF _x liquidus						
1053 (780)	Fe ₂ O ₃ o.	in air	1173 (900)	1 + 3	liquid PF _{1±x}	77.7 56.4
1073 (800)	Fe ₂ O ₃ o.	in air	1173 (900)	0.5 + 2	liquid PF _{1±x}	74.9 54.7
1073 (800)	Au o.	in air	no	72	liquid PF _{1±x}	75.3 53.7
1098 (825)	Au o.	in air	no	72	liquid PF _{1±x}	71.7 49.3
1123 (850)	Fe ₂ O ₃ o.	in air	1173 (900)	0.2 + 1	liquid PF _{1±x}	67.2 50.0
1148 (875)	Fe ₂ O ₃ o.	in air	1173 (900)	0.5 + 2	Liquid P _{1+x} F _{12-x}	66.0 11.5
1148 (875)	Au o.	in air	no	24	liquid P _{1+x} F _{12-x}	65.3 8.2
1173 (900)	Fe ₂ O ₃ o.	in air	1273 (1000)	0.5 + 1	liquid P _{1+x} F _{12-x}	64.0 9.4
1223 (950)	Fe ₂ O ₃ o.	in air	1253 (980)	0.2 + 1	liquid P _{1+x} F _{12-x}	61.5 10.3
1273 (1000)	Pt-Rh wire	in air	1303 (1030)	0.5 + 2	liquid P _{1+x} F _{12-x}	58.2 8.5
1323 (1050)	Pt-Rh wire	in air	1343 (1070)	3	liquid P _{1+x} F _{12-x}	55.2 8.8
1373 (1100)	Pt-Rh wire	in air	1393 (1120)	2	liquid P _{1+x} F _{12-x}	52.7 8.6

o.—open basket, *—the equilibrated areas of the phase were too small to analyze. Times are listed as (premelt time) + (equilibration time).

important for the slag + spinel + magnetoplumbite equilibrium results:

1. A range of solutions was observed for the magnetoplumbite $P_{1+x}F_{12-x}$ (see Figure 4), with the Fe:Pb ratio gradually increasing from 5.5 at 1093 K (820 °C) to 9 at 1373 K (1100 °C). The composition of magnetoplumbite $P_{1+x}F_{12-x}$ in equilibrium with slag and spinel is suggested to follow the line between $PbFe_{12}O_{19}$ and $Pb_2Fe_{11}O_{18}$ ($Pb_{1+x}Fe_{12-x}O_{19-x}$), according to the mechanism of substitution [1] suggested by [39] Analogous substitution of $2Fe^{3+}$ by ($Zn^{2+} + Pb^{2+}$), along the join between $PbFe_{12}O_{19}$ and $Pb_2ZnFe_{10}O_{18}$ ($Pb_{1+x}Zn_xFe_{12-2x}O_{19-x}$), was observed in the experiments by Jak *et al.* [8–10] in the multicomponent Pb-Zn-Fe-Ca-Si-O system in air (Figure 7).

Although the oxygen content was not measured directly with EPMA, some information can be obtained from the total oxide wt pct. The Fe concentrations were recalculated to Fe_2O_3 ($FeO_{1.5}$) so that the excess over 100 pct relative to the $PbO-FeO_{1.5}$ join is related to the ferrous iron in a phase. The trend of the ferrous iron estimated this way for the magnetoplumbite ($P_{1+x}F_{12-x}$) phase agrees with the substitution mechanism $2Fe^{3+} + O^{2-} \rightarrow Pb^{2+} + Fe^{2+}$ [39] that is another indication supporting the selected formula for this phase ($Pb_{1+x}Fe_{12-x}O_{19-x}$, $0 < x < 1$).

2. The solubility of PbO in spinel was found to be higher in the presence of $P_{1+x}F_{12-x}$ (up to 3 mol pct on the Pb/(Pb + Fe) atomic ratio), compared to the experiments in equilibrium with Pb metal. The temperature dependence of lead oxide solubility in spinel was small, being

Table II. Composition Ranges and Structure of Phases in the PbO-”FeO_x” System

Phase name	Formula	Abbreviation	Range of Fe:Pb Ratios	Crystal Structure
Magnetoplumbite	$Pb_{1+x}Fe_{12-x}O_{19-x}$, $0 < x < 1$	$P_{1+x}F_{12-x}$	5.5 to 12	hexagonal
Plumboferrite	$PbFe_{5+x}O_{8.5+1.5x}$, $0 < x < 1$	PF_{5+x}	5 to 6	hexagonal
1:1 Lead ferrite	$Pb_2Fe_{2\pm x}O_{5\pm 1.5x}$, $0 < x < 0.2$	$PF_{1\pm x}$	0.9 to 1.1	tetragonal

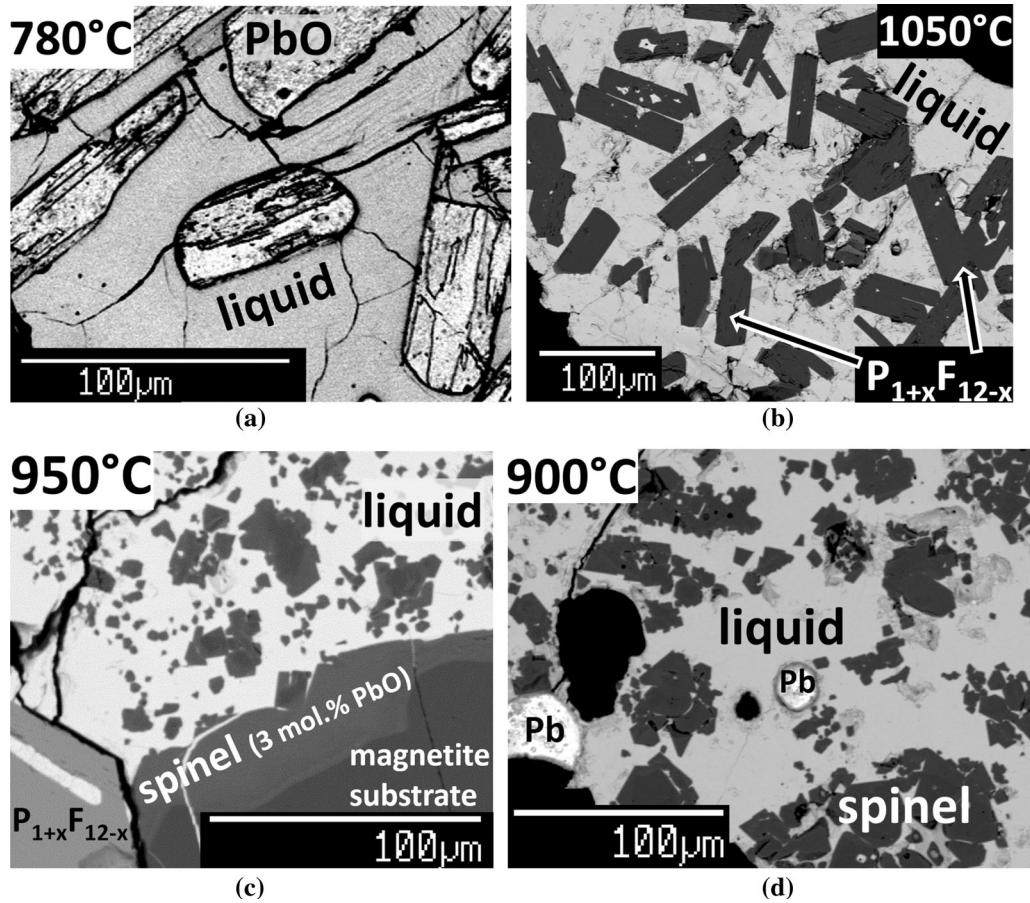
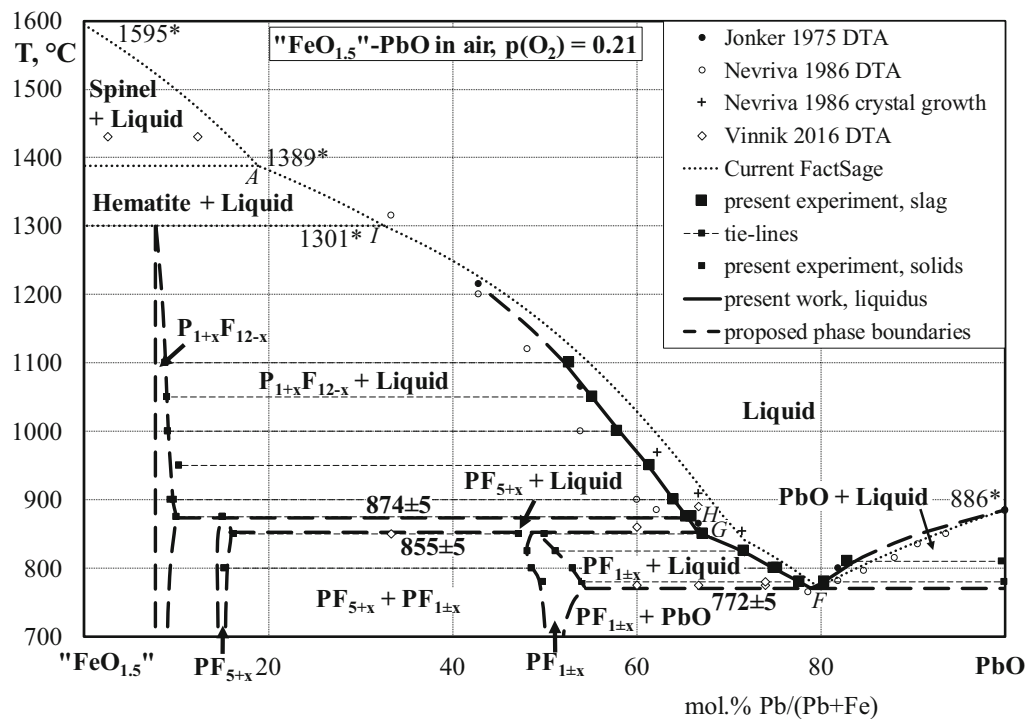
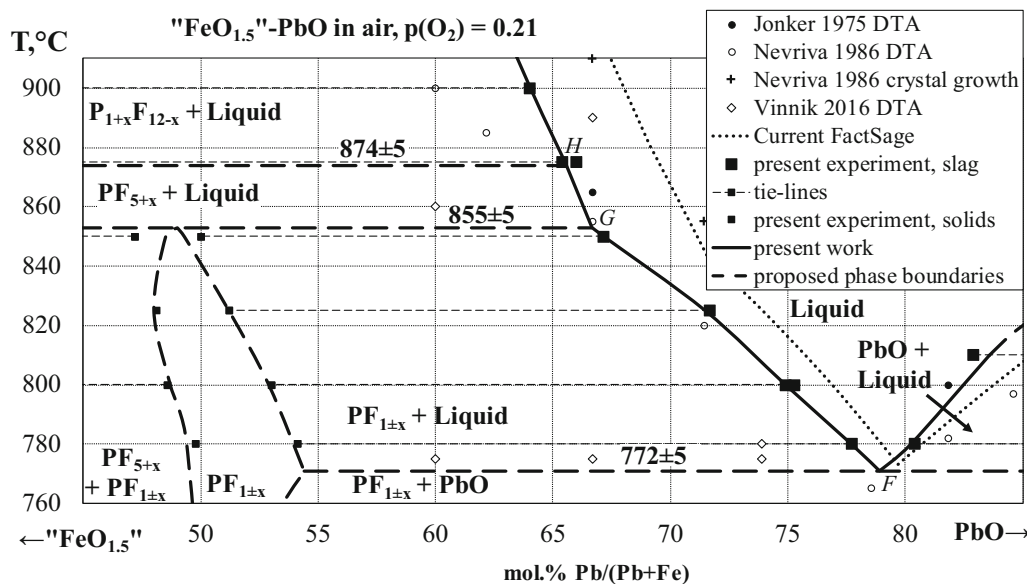


Fig. 1—Backscattered scanning electron micrographs of quenched slag in the Pb-Fe-O system in equilibrium with one or two crystalline phases or metal at various $p(O_2)$: (a) Liq + PbO + air, (b) Liq + $P_{1+x}F_{12-x}$ + air, (c) Liq + $P_{1+x}F_{12-x}$ + spinel (in Ar), (d) Liq + spinel + Pb metal (in Ar).



(a)



(b)

Fig. 2—Proposed phase diagram for the “FeO_{1.5}”-PbO system in air ($p(\text{O}_2) = 0.21$) showing experimentally measured liquid and solid compositions, compared with FactSage predictions (temperatures with asterisk *) and the literature data: Jonker 1975,^[24] Nevrya 1986,^[27] Vinnik 2016.^[31] (a) Complete diagram; (b) enlarged part of low-temperature liquid–solid equilibria.

slightly lower (2.4 to 2.7 mol pct) at 1323 K to 1373 K (1050 °C to 1100 °C), compared to 2.5 to 3.0 mol pct at 1093 K to 1173 K (820 °C to 900 °C).

D. Ternary Phase Diagrams

The phase compositions combined with thermodynamic modeling of the $\text{Fe}^{2+}/\text{Fe}^{3+}$ ratios in the liquid

phase were used to construct a ternary liquidus diagram (Figure 5) and a tentative 1048 K (875 °C) isothermal section (Figure 6) that represent the phase equilibria in the PbO-FeO_{1.5}-FeO system over a wide range of conditions. Figure 5 shows that only a small part of the PbO-FeO_{1.5}-FeO liquidus could be covered by experiments at or below 1373 K (1100 °C), and the results correspond to low concentrations of ferrous iron in the liquid.

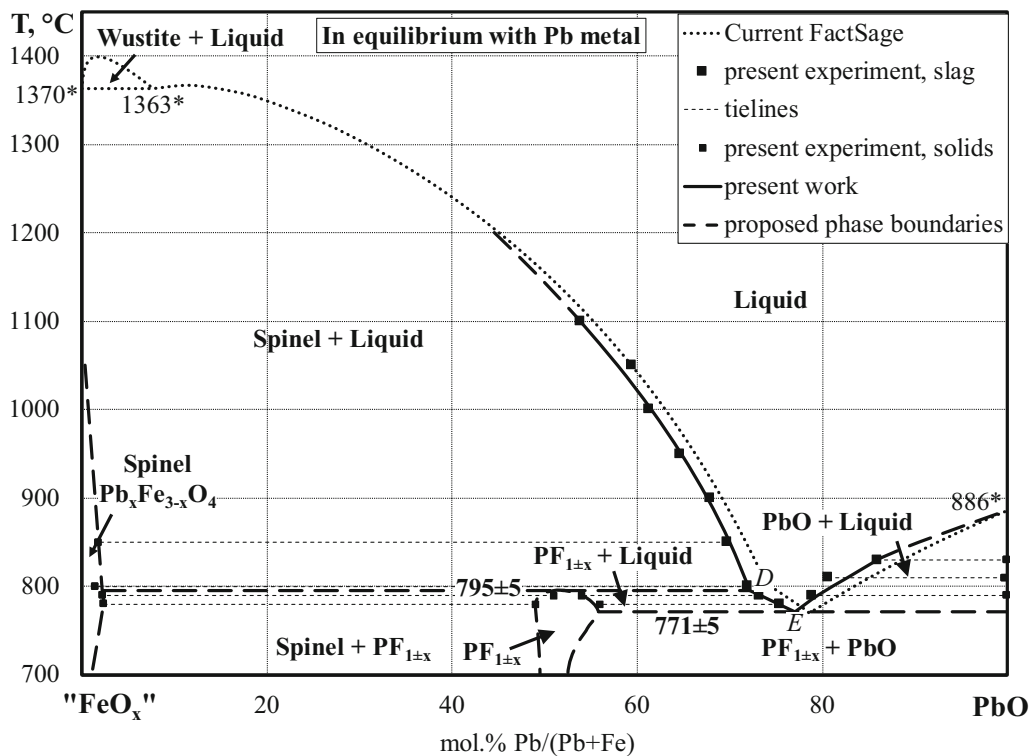


Fig. 3—Proposed phase diagram for the “FeO_x”-PbO system in equilibrium with metallic Pb, compared with FactSage predictions (temperatures with asterisk *).

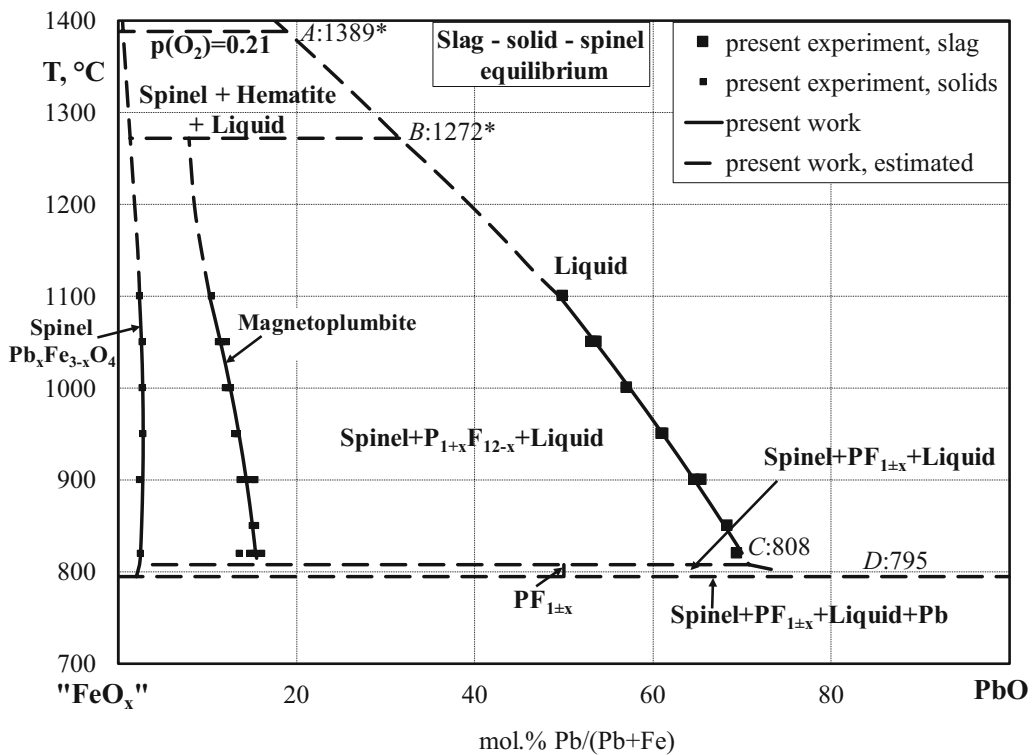


Fig. 4—Proposed phase diagram for the “FeO_x”-PbO system for the section of (solid oxide) + (spinel) + (liquid oxide) equilibrium, combined with FactSage predictions (temperatures with asterisk *).

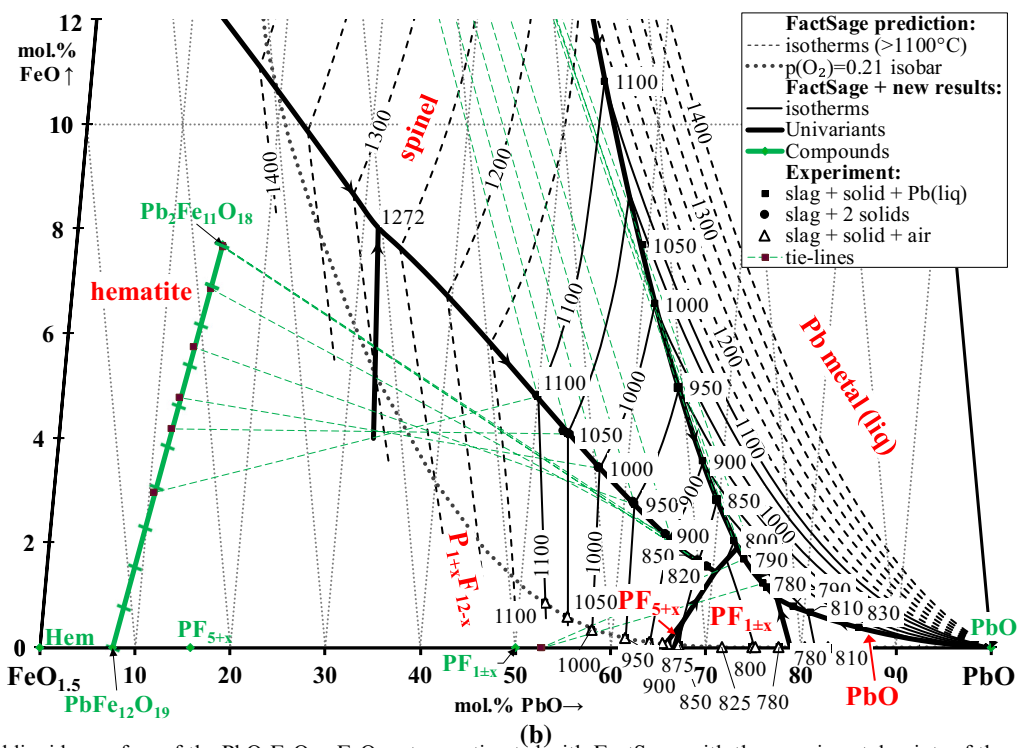
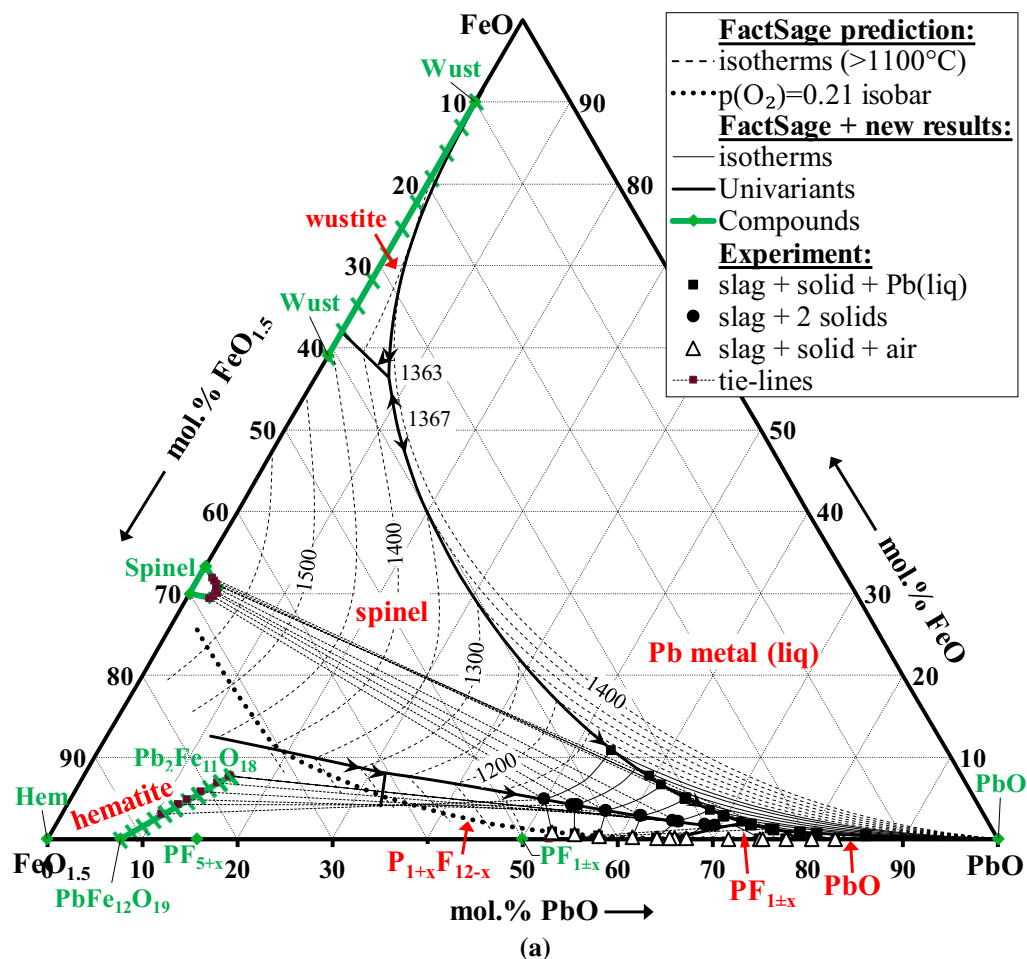


Fig. 5—Proposed liquidus surface of the PbO-FeO_{1.5}-FeO system, estimated with FactSage, with the experimental points of the present work. Note that the $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ ratios were not measured, so the position of the points is adjusted to the model curves. (a) Complete diagram; (b) detail at low FeO concentrations; (b') same as (b) without experimental points; (c) detail to show the low-temperature phase fields of $\text{PF}_{1\pm x}$ and PF_{5+x} .

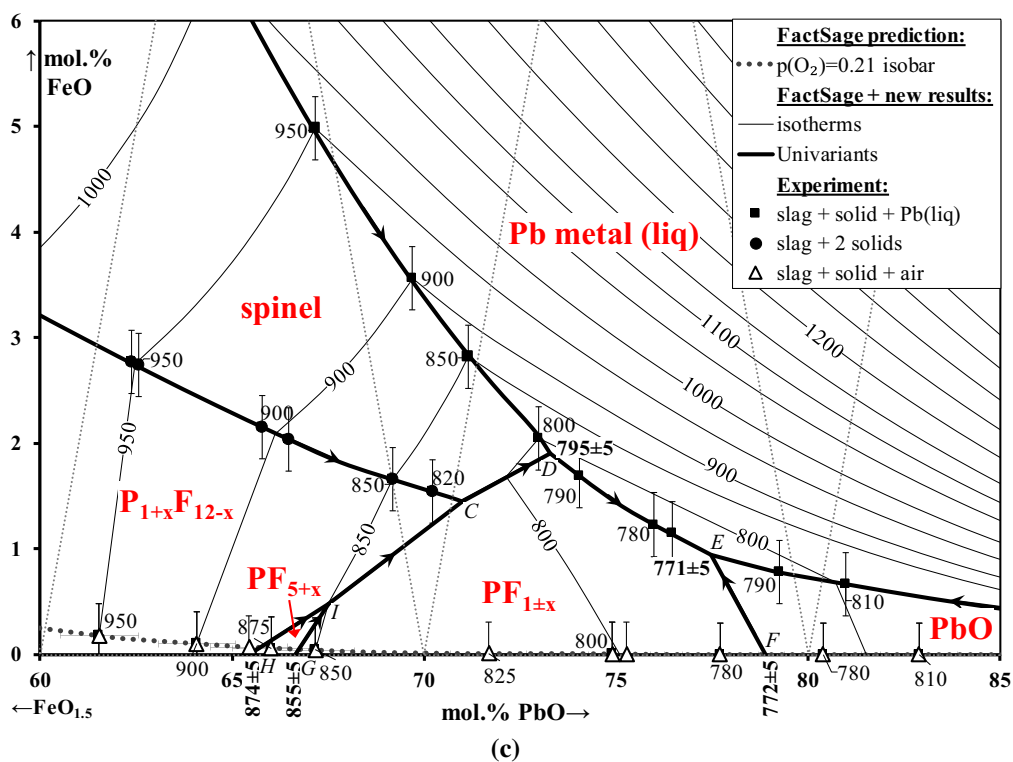
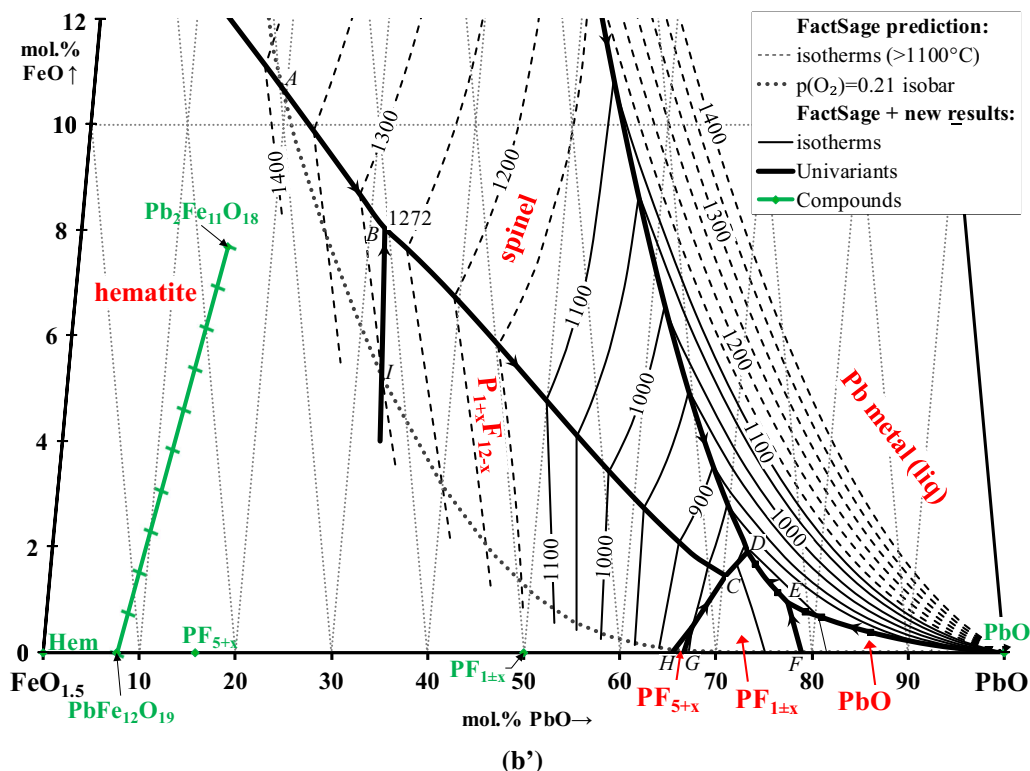


Fig. 5—continued.

Most of the PbO-FeO_{1.5}-FeO compositional triangle belongs to the Pb metal and spinel primary phase fields. Limited ranges at high temperatures are occupied by wustite and hematite primary phase fields. The reactions hematite + Liquid → spinel + magnetoplumbite and Liquid → wustite + spinel + Pb are

estimated to occur at 1545 K (1272 °C, point *B*) and 1636 K (1363 °C), respectively, according to the current FactSage database. The part of the diagram close to the PbO-“FeO_{1.5}” join at high temperatures is not shown, since it corresponds to high oxygen partial pressures, >1 atm.

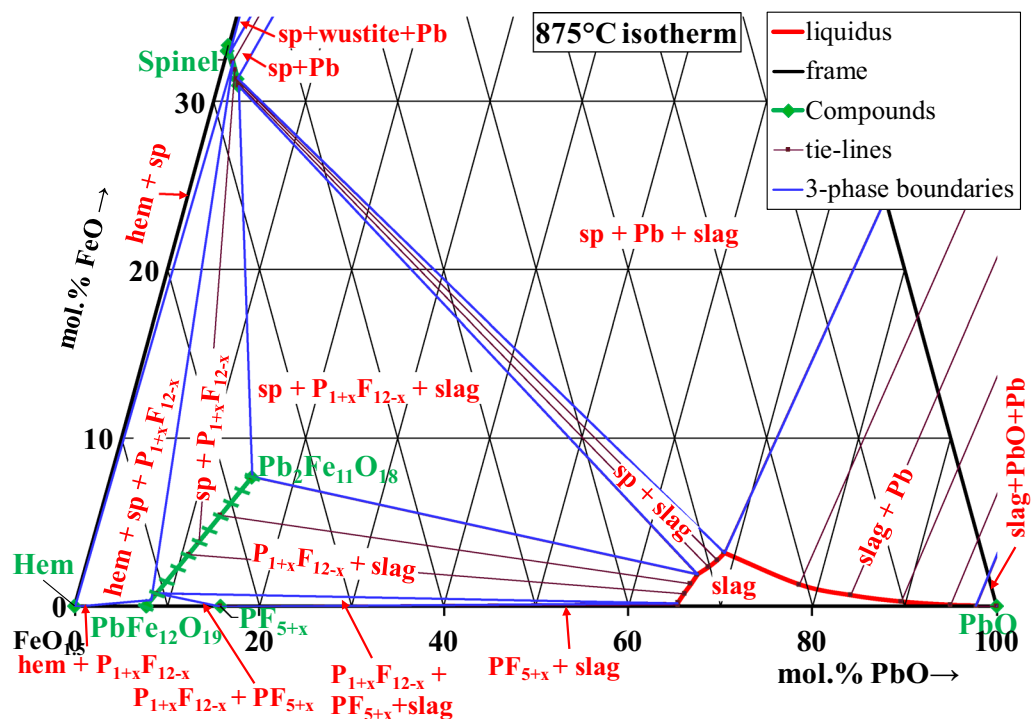


Fig. 6—Tentative isothermal section of the PbO-FeO_{1.5}-FeO phase diagram at 1048 K (875 °C). Abbreviations: P_{1+x}F_{12-x} = magnetoplumbite (Pb_{1+x}Fe_{12-x}O_{19-x}, 0 < x < 1), PF_{5+x} = plumbiferite, sp = spinel (Pb_xFe_{3-x}O₄, 0 < x < 0.1), hem = hematite.

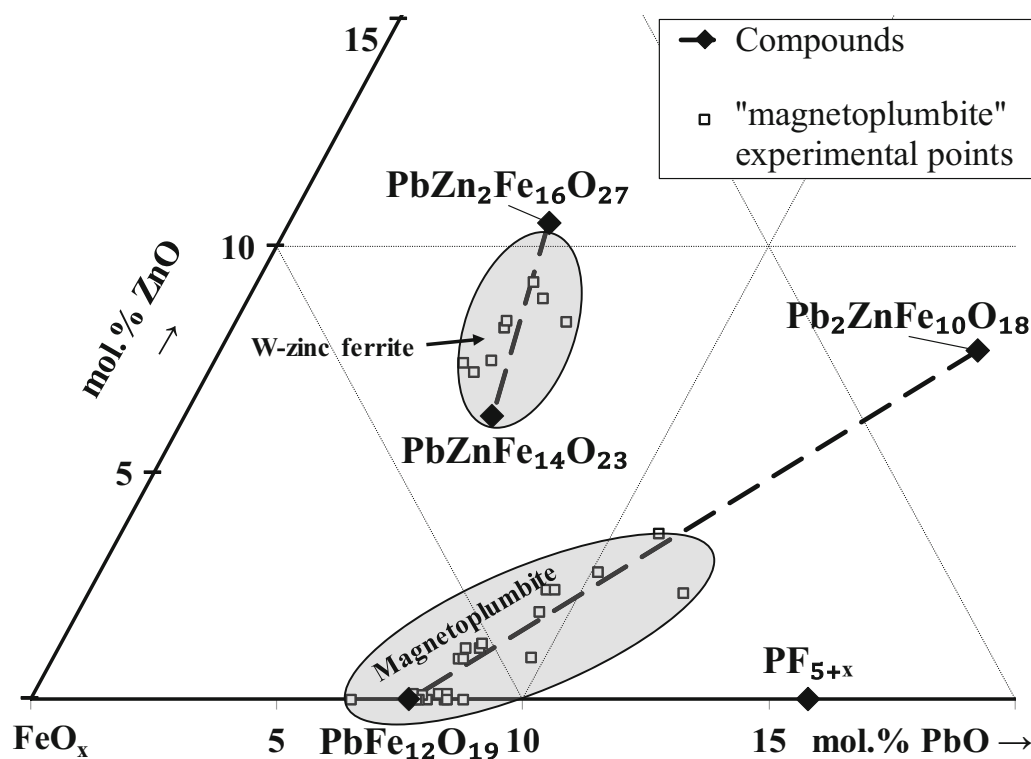
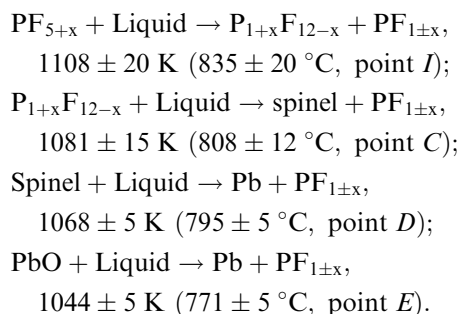


Fig. 7—Compositions of the phases identified as “magnetoplumbite” in the experiments by Jak *et al.*^[8–10] in the multicomponent Pb-Fe-Ca-Zn-Si-O system in air, projected onto the PbO-“FeO_x”-ZnO triangle (the average molar fractions of CaO and SiO₂ are only 0.5 and 0.3 mol pct, respectively). Note that in addition to the actual magnetoplumbite solid solution along the PbFe₁₂O₁₉-Pb₂ZnFe₁₀O₁₈ join, another separate phase with approximate composition PbZnFe₁₄O₂₃-PbZn₂Fe₁₆O₂₇ can be identified, corresponding to W-type hexaferrite (Haberey^[45]).

At low temperatures, the following invariant reactions are suggested:



IV. CONCLUSIONS

New phase equilibria information in the Pb-Fe-O system has been obtained in air, in equilibrium with metallic lead, and at intermediate oxygen potentials. The equilibria at reducing conditions (with Pb metal or spinel) have been studied for the first time, since all previous authors had investigated this system in air or oxygen atmosphere. The studied range of temperatures covered 1048 K to 1373 K (770 °C to 1100 °C), and involved the equilibria of slag with spinel, lead ferrites ($\text{PF}_{1\pm x}$, PF_{5+x} and $\text{P}_{1+x}\text{F}_{12-x}$), and lead oxide. Limited ranges of solid solutions in PbO, $\text{PF}_{1\pm x}$, and spinel were detected. A range of solid solutions with Fe:Pb ratios between 5 and 11 was observed in $\text{P}_{1+x}\text{F}_{12-x}$, and Fe:Pb between 5 and 6 in PF_{5+x} . The phase equilibria information provided by the current study will be used for complete revision of the existing thermodynamic database of lead-containing slag system (Pb-Zn-Fe-Cu-Si-Ca-Al-Mg-O).

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